2016-17
2017-18
2018-19

Murad Sir Part

previous question solve

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Date :......./......./.........

System Analysis

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(13) How many symbols are used for Data Flow Diagram? (8)

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(18) How report ~~different~~ from different persons and places? (9)

(19) What are the cases where illegal data flows happen? (8)

(20) What are the significance of database integrity in system analysis and Design? (

(21) ~~How many symbols are used for Data Flow Diagram?~~

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(21) Give an outline for a standard oral presentation. (9)

(22) Make comparison between factual and administrative format for written report. (9)

~~(24)~~

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1) Body languages:

Body language is a nonverbal communication.

By research:

1. Verbally – 7% (in word)
2. Tone of voice – 38%
3. Facial and body expression – 55%

- If you hear only word you will miss what you ~~who~~ want to say.

There are three aspects of body language:

1. **Facial disclosure:** (प्रकाशन)
   one of the most controlled part of the body.
2. **Eye contact:**
   Direct eye contact can cause strong feelings either positive or negative.
3. **posture (attitude):** (मुद्रा)
   Least controlled aspect of the body.

(2) Proxemics:

Proxemics is the study of how people use space when they are talking to each other.
It helps us understand personal space during conversations.

Example:
Imagine you're talking to a friend. If you stand very close, like less than 1 foot away, your friend might feel uncomfortable because it's too close for normal conversation. However, if you're far away, like 10 feet, it might feel like you're not engaged or distant in the conversation.

(3) Spatial Zones:

Zone	around Space
Intimate zone	closer than 1.5 feet
personal zone	from 1.5 feet to 4 feet
social zone	from 4 feet to 12 feet
public zone	beyond 12 feet

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4. Describe different types of questionnaires:

A questionnaire is a set of written questions used to collect information from people.
It is often used in research, surveys, and information systems to understand people's opinions, experiences, or needs.

and opinions

Free format questionnaires:

Space
greater latitude in the

They give the person more freedom to answer.
A question is asked, and the person writes the answer in the blank space given after the question.

Question: What do you like to do in your free time?
Answer: I like to read books and play football with my friends.

Fixed format questionnaires:

require selection of

Fixed format questionnaires give the person choices to select from.
The person cannot write their own answer.
They must choose one answer from the given options.
This type is easy to answer and easy to check or count.

- Types of fixed format questions:
  - → Multiple-choice questions
  - → Rating questions
  - → Ranking questions

(5) What are the advantages and disadvantages of questionnaires?

Advantages:

quickly

Advantages:
1.Quick to get answers from many people.
2.Low cost, as you don’t need to spend much money.
3.Easy to compare answers from everyone.
4.Anonymous responses, so people can be honest.
5.Can be sent to many people, like online, by email, or on paper.

analyzed quickly.

Disadvantages:

(i) Respondents often low

Disadvantages:
1.Limited answers (e.g., in multiple-choice, there are only a few options).
2.Low response rate, meaning not everyone may answer.
3.People may misunderstand the questions.
4.Some answers may not be detailed enough.
5.There is no one to help if someone doesn’t understand the question.

(iv) Not possible to read body language.

(v) Good questions are difficult to prepare.

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⑥ What are the advantages and disadvantages of interviews?

Interviews are a fact finding technique whereby the systems analysts collect information from individuals through face to face interaction.

Advantages:

(i) opportunity to motivate the interviewee.
(ii) More feedback from the interviewee.
(iii) permit system analyst to adapt or reward questions for each individual.
(iv) possible to read body language.

Disadvantages:

(i) Time consuming.
(ii) Success highly depends on system analysts human relation skill.
(iii) May be impractical due to location.

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⑦ What should do and avoid for interview session?

Do	Don't (Avoid)
1. Be polite	1. continuing an interview unnecessarily
2. Listen carefully	2. Assuming an answer is finished.
3. Maintain control	3. Revealing (প্রকাশ করা) verbal and nonverbal clues.
4. Inquiry	4. Revealing your personal biases (পক্ষপাত).
5. observe mannerisms and nonverbal communication	5. Talking instead of listening.
6. Be patient	6. Assuming anything about the topic and the interviewee.
7. Keep interviewee at compont	7. Tap recording – a sign of poor listening skills.
8. Maintain self control.	

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(8) Brainstorming :

Brainstorming is a technique for generating ideas during group meetings. participants are encouraged to generate as many ideas as possible in a short period of time without any ~~any~~ analysis untill all the ideas have been exhausted.

Brainstorming guideline :

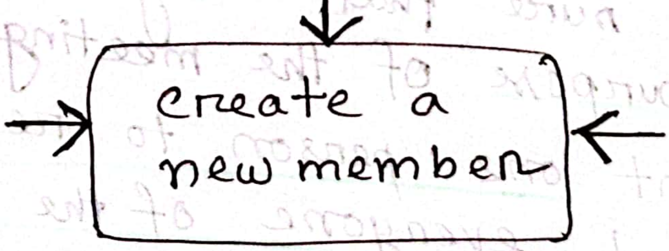
1. Isolate the appropriate people in a place that will be free from distructions and interruptions.
2. Make sure that everyone understands the purpose of the meeting.
3. Appoint one person to record ideas.
4. Remind everyone of the brainstorming rules
5. Within a specified time period, team members call out their ideas as quickly as they can think of them.
6. After all ideas have been recorded only then idas be analyzed and evaluated.
7. Refine, combine and improve the ideas that were generated earlier.

⑨ What are the most common process errors occur when a data flow diagram are drawn for a system?

▶ A data flow diagram is tool that depicts the flow of data through a system and the work or processing by that system.

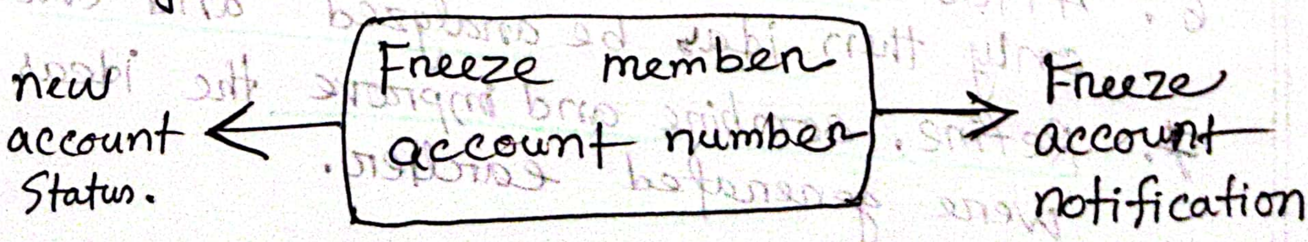
(I) <u>Input but no outputs:</u>

process has inputs but no outputs. It is called black hole. because and then data enter the process disappear.



(II) <u>process has outputs but no inputs:</u>

Unless you are 'David Copperfield' it's a miracle. In most cases input flows were likely forgotten.



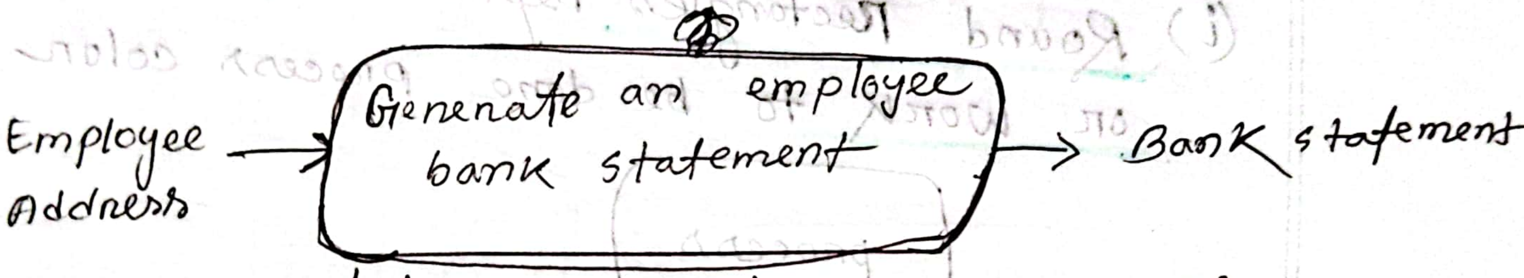
III Inputs are innsufficient to produce the Output:

It is called greay hole because

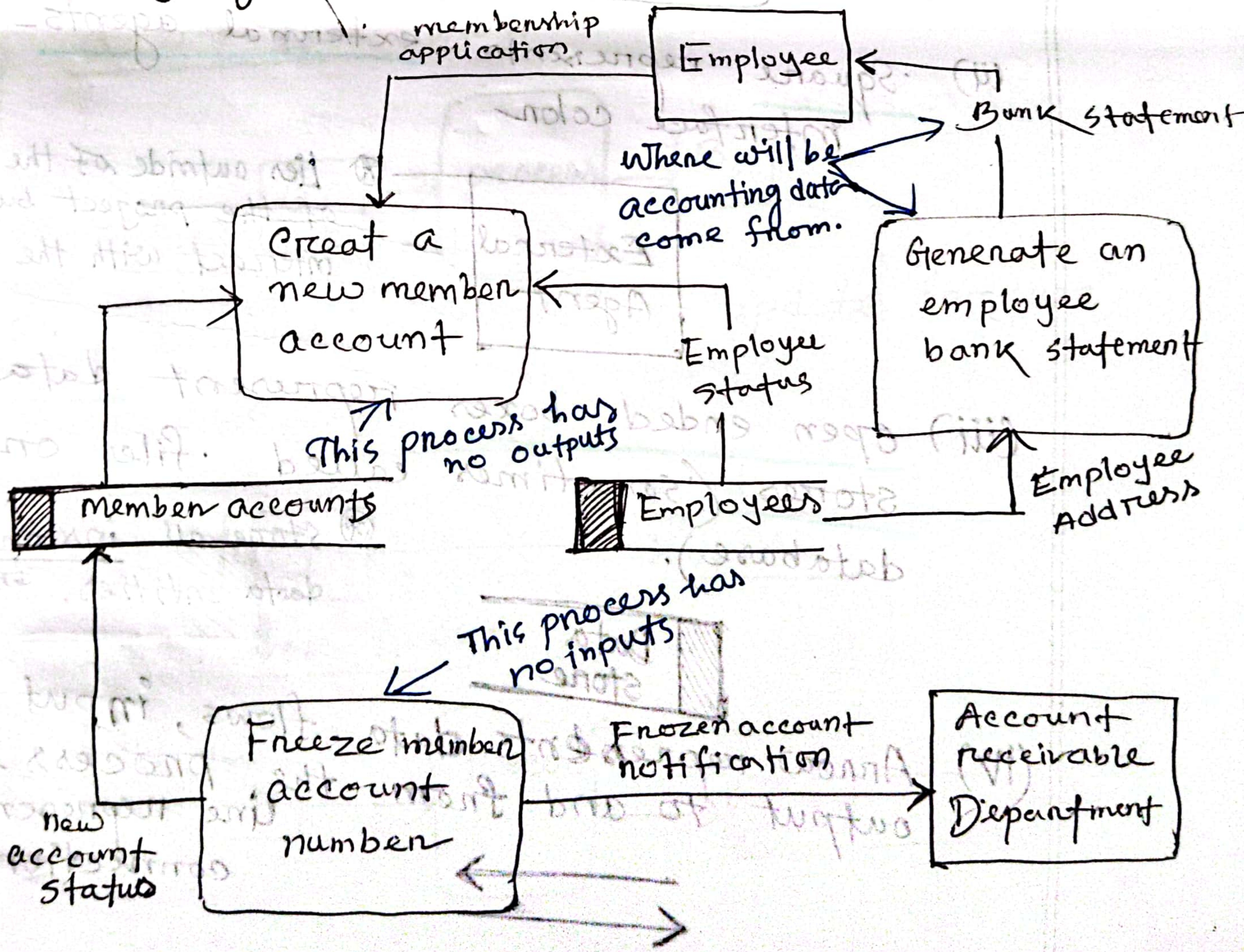
→ Misnamed process.

→ misnamed inputs and outputs.

→ Incomplete facts.



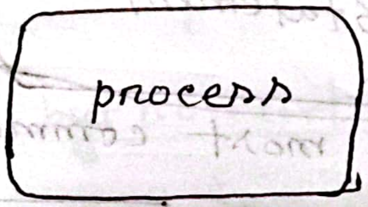
• gray hole are most common errors.



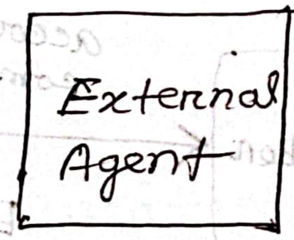
(10) How many symbols are used for Data Flow Diagram?

- Three symbols and one connection are used for Data flow Diagram.

(i) Round Rectangles represent processes or work to be done process colon



(ii) Square represents external agents – interface colon

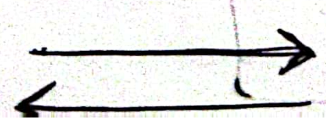


* Lies outside of the scope সুযোগ of the project but that interact with the system.

(iii) Open ended boxes represent data stores (sometimes called files or database).

* Store all instances উদাহরণ of data entities.

(IV) Arrow represents data flows, input and output to and from the process. and line represent connection

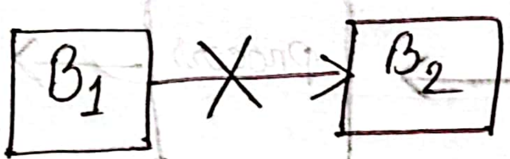


(11) What are the cases where illegal data flows happen?

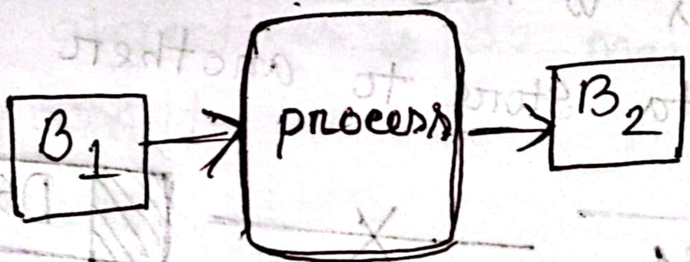
A

Data flow analysis is that all flows must begin with or end at a processing step.

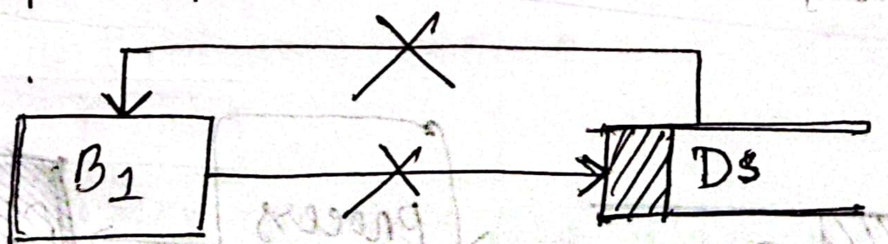
(1) A process is need to exchange data flows between external agents.



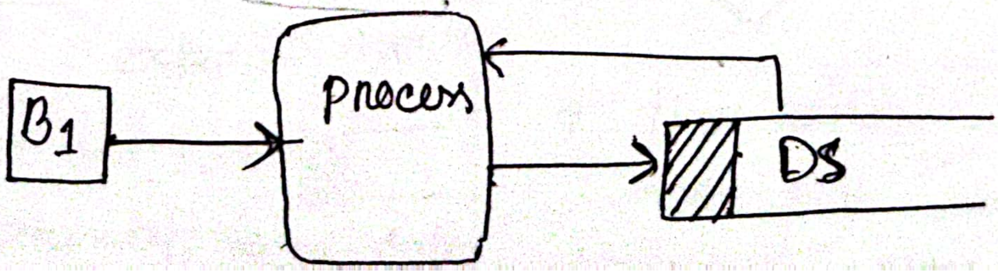
Correct data flows:



(2) A process is needed to update (or use) a data store.



Correct data flows:



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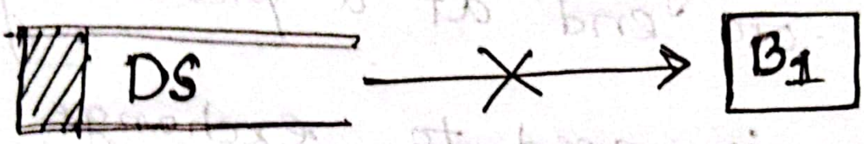
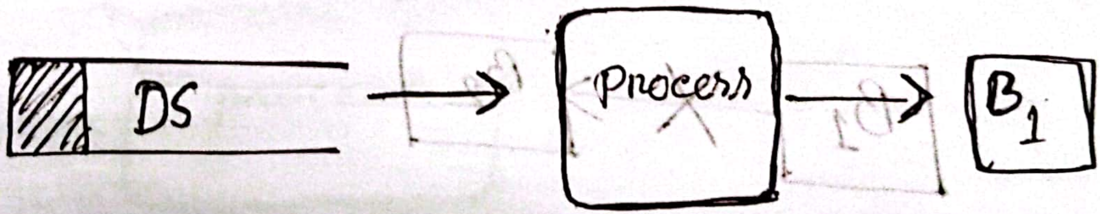
③ A process is needed to present data from a data store.

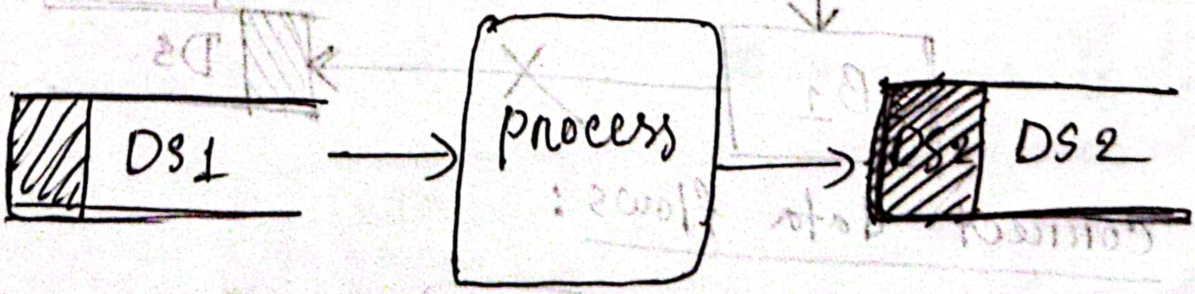


Correct data flows:



④ A process is needed to move data from one data store to another

Correct data flows:



(12) ~~Make~~ Make Comparison between DFD and ERD

DFD	ERD
1. It stands for Data Flow Diagram.	1. It stands for Entity Relationship Diagram.
2. DFD shows the flow of Data through a system.	2. ERD shows the relationship between entities.
3. main objectives is to represent the process and data flow between them.	3. Main objective is to represent data object.
4. Symbol of DFD: * round recamtangle represent process * Square represent external Agents * open end box represent stone data * Arrow → represent flow of data	4. Symbol of ERD: * rectangles represent entity. * diamond box represent relationship * line and Standard notation represent cardinality.

(13) A dollar today is worth more than a dollar one year from now – What is the significance of the statement. Explain with suitable example.

A dollar today is worth more than a dollar one year from now. You could invest that dollar today and through accrued interest, have more than one dollar a year from now. Thus you'd rather have that dollar today than in one year later.

সঞ্চিত

Suppose we are going to realize a benefit of \$20,000 two years from now. What is the current dollar value of that \$20,000 benefit?

The current value of the benefit is the amount of money we would need to invest today to have \$20,000 two years from now. If the current return on investments is running about 10 percent, Date :....../....../......

$$PV = \frac{\$20{,}000}{\left(1+\frac{10}{100}\right)^2}$$

$$= \$16528$$

investment of \$16528 today ; would give us our \$20,000 in two years.

Another example, suppose we have option of receiving \$1000 now and one years from now. Today \$1000 has more value and utility then it will two years from now. because of / due to the opportunity cost associated with the delay.

Significance of the Statement:

1. opportunity cost.

2. Time value of money.

3. Inflation consideration.

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(14) Give an outline for standard oral presentation.

1. Introduction (one-sixth of total time available)
   A. problem statement :
   B. work completed to date

2. part of the presentation (two thirds of total time available)
   A. Summary of existing problems and limitations.
   B. Summary description of the proposed system.
   C. Feasibility analysis.
   D. proposed schedule to complete project.

3. Questions and concerns from the audience :

   There is no time allocation. It is determined by those, asking the questions and voicing their concerns.

4. Conclusion: (one-sixth of total time available)
   A. Summary of proposal.
   B. Call to action (request for whatever authority you require to continue system development.

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(15) Make comparison between factual and administrative format for written report.

Factual Format	Administrative Format
1. Introduction	1. Introduction.
2. Methods and procedures	2. Conclusions and Recommendations.
3. Facts and details	3. Summary and discussion of facts and details
4. Discussion and analysis of facts and details.	4. Methods and procedures.
5. Recommendations	5. Final conclusion
6. Conclusion	6. Appendices with facts and details

Date :......./......./.........

(16) Difference between database and conventional file.

File System	Database
1. A process that manages how and where data on a storage disk is stored. Accessed and managed	1. An organized collection of data that can be easily accessed, managed and updated.
2. Has high data inconsistency.	2. Maintain data consistency
3. Structure is simple.	3. Structure is complex
4. Data sharing is hard.	4. Data sharing is easy.
5. There is high redundancy	5. There is low redundancy
6. Not very secure.	6. More secure.
7. No backup and recovery process.	7. There is backup recovery.

(17) What is present value of money?

→ ~~Today's~~ Present value of tomorrow's cash

▸ Future Value:

→ Tommorrows value of today's cash.

▸ (The current value of a future sum of money.)

→ (Present value, a concept based on time value of money, States that a sum of money today is worth much more than the same ~~sum~~ sum of money in the future) and is calculated by dividing the future cash flow by one plus the discount rate raise to the number of periods.

▸ $PV = 10{,}000$ tK

$r = 20\% = 0.2$

$n = 5$

We know,

$$PV = \frac{FV}{(1+r)^n}$$

$$\therefore FV = PV \times (1+r)^n$$

$$= 10{,}000 \times (1+0.2)^5$$

$$= 24883.2$$

• Example

$$\text{present value} = \frac{\text{Future value}}{(1+i)^n}$$

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কিছু নাই

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(18) How can we calculate present value of money?

From online

present value is calculated by dividing the future cash by one plus discount rate raise to the number of periods.

$$PV = FV \frac{1}{(1+i)^n}$$

Step-1 : put expected Future value of the investment in a formula.

Step-2 : put expected rate of return on your investment

Step-3 : Number of period you are Investing.

Where,

$n \rightarrow$ number of years

$i \rightarrow$ discount rate

$FV \rightarrow$ Future value of money.

If the future cash flow is 1 unit of currency, the formula simplified to

$$PV_n = \frac{1}{(1+i)^n}$$

+ ~~আগের~~ Page এর example লিখো।

(19) The time value of money is not taken into account for payback analysis – Explain the statement with appropriate example.

Payback analysis is a simple and popular method for determining when an investment will pay for itself.

Payback period is refer to the period of time takes to repay the sum of the original investment.

Payback period has some significant drawback.

1. Fails to take into account the time value of money.
2. Fails to adjust the cash in flow.

For example:

A $1000 investment which return $500 per year.

- Initial investment $1000
- annual cash flow $500
- payback period = $\frac{\text{Initial investment}}{\text{annual cash flow}}$

$$= \frac{\$1000}{\$500} = 2 \text{ years}$$

However, payback analysis is that it doesn't consider the time value of money. In reality money has the potential to earn returns or interest over time.

In summary, payback analysis is a straighforwadrd method to assess the recovery time of an investment but doesn't consider the time value of money.

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20) What are the most important skill that a system analyst must have ?

1. Technical knowledge
2. Analytical skills
3. Communication skills
4. project management skills.
5. problem solving skills
6. Attention to detail
7. Adaptability
8. collaboration
9. Business Knowledge
10. Critical thinking
11. creativity
12. Interpersonal skills.
13. Leadership

প্রতিটার 2/১ লাইনে ব্যাখ্যা

(21) How Report differ from different persons and places?

The written reports is the most used method by the analyst to communicate with system user.

There is a tendency to generate large reports that look impressive but it is not effective. Sometimes reports are necessary but often they are not. If a manager receives 500 page technical reports, the manager may skip it.

Length of report for different person or place:

1. To executive level manager – one or two page report
2. To mid level manager – three to five page
3. To supervisory level manager – less than 10 page
4. To clerk level manager – less than 150 page.

Different format for different places:

For documentation

Factual Format	Administrative Format
Ques NO-15 এর অনুরূপ	

(22) What are the Significance of Database integrity in system analysis and Design?

1. Data accuracy and Reliability
2. Consistency
3. Data validity
4. Referential integrity
5. Avoiding Data Redundancy
6. Data integrity validation
7. Data maintainance
8. Reporting & analysis
9. Data migration and integration

[2/3 লাইন লেখে দিব]

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How can we calculate present value of money:

The Current value actually called the present value, of a dollar at any time in the future can be calculated using the following formula:

$$PV_n = \frac{1}{(1+i)^n}$$

Where PV_n is the present value of \$1.00 n years from now and i is the discount rate.

Given, discount rate of a company is 12 percent. Therefore the present value of a dollar at any time in the future can be calculated using the following formula:

Here, $n=2$

$$PV_2 = \frac{1}{(1+1.2)^2}$$

$$= 0.797$$

From Google

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We ~~stal~~ know that, a dollar today is worth more than a dollar a year from now. But it looks as if, it is worth less. This is an illusion.

If you have 79.7 cents today, it is better than having 79.7 cents two years from now.

——o——